

Cycles 1-2 Clone 2 Report — M-DAW-SPIKE-1/vst3-render-nondeterminism-characterization (Fork 87da4f517029)

2026-08-29

Contents

Cycles 1-2 Clone 2 Report — M-DAW-SPIKE-1/vst3-render-nondeterminism-characterization (Fork 87da4f517029)	1
Abstract	1
Verdict	1
Rubric SHA Anchor Chain (Three-Way Byte-Equal)	2
Method	2
Cycle Disposition	2
Test Surface	2
Anchor Preservation	3
Ledger Events (9 Shadow Rows Under -clone-2 Suffix)	3
MODERATE Findings (Handoff to c37, Not Corrections Owed This Branch)	3
Anti-Patterns Locked (6-Count Stable Including c35 A #6)	4
State-Machine Discipline (c29 Lemma Respected)	4
Merge Disposition	4
Standing Constraints (Unchanged)	4
Cycle-37 Handoff (Enumerated in Merge Report + Audit)	4
Cycle-Bounding for c37 palette-v3 Decision	5
Cumulative Progress	5

[OUTPUT: report_cycles_1-2_clone_2]

Cycles 1-2 Clone 2 Report — M-DAW-SPIKE-1/vst3-render-nondeterminism-characterization (Fork 87da4f517029)

Abstract

Cycles 1-2 of clone-2 (fork 87da4f517029) close the cycle-36 Branch C auditor-carried alternate candidate M-DAW-SPIKE-1/vst3-render-nondeterminism-characterization at **MIXED** with per-plugin labels preserved. Cycle 1 executed the full N=5-render analytical probe on Surge XT + Dexed VST3 under identical BLAS pins and torch seed 0; per-plugin classification yielded Surge XT = STRUCTURAL and Dexed = SMALL (byte-adjacent-tolerant at float32 epsilon), aggregated as MIXED under the frozen 3-verdict rubric. Cycle 2 is c30-codified verification-only standby (honest low-output memo; zero writes; auditor decision **COMPLETE**). Three MODERATE findings are cleanly handed to c37; c31 STILL_GAP and c35 A anti-patterns are not re-opened.

Verdict

MIXED (VALIDATED at cycle 1 per rubric aggregation; **COMPLETE** at cycle 2 standby).

Per-plugin labels preserved in verdict.json for c37 freedom:

Plugin	Label	Load-Bearing Numbers
Surge XT	STRUCTURAL	max pairwise RMS $\approx$ 0.098 (pair 1-3); max_abs_sample $\approx$ 0.327 (pair 2-4)
Dexed	SMALL	byte-adjacent at float32 epsilon

Rubric SHA Anchor Chain (Three-Way Byte-Equal)

Location	SHA-256
docs/vst3_nondeterminism_characterization_rubric.md	ddc70837...d9bf560
data/vst3_nondeterminism/rubric_hash.txt	ddc70837...d9bf560
verdict.json.rubric_hash	ddc70837...d9bf560

Verified byte-equal at cycle 1; unchanged at cycle 2.

Method

Loaded each of Surge XT and Dexed VST3 in isolation via DawDreamer; hydrated via c33 P1 set_parameter(i, v) iteration (100% param coverage: 2855/2855 Surge XT; 2238/2238 Dexed). Rendered the c31 fixed 8 s @ 44.1 kHz stereo ascending-diatonic MIDI N=5 times per plugin in per-run isolated tempfile.mkdtemp() dirs under identical single-thread BLAS pins (OMP=MKL=OPENBLAS=1) + torch seed 0.

Computed per-plugin: SHA-256 per render; pairwise RMS across all $C(5,2)=10$ render pairs; per-pair envelope correlation (Pearson on RMS envelopes at hop=512); per-pair mel_l1_db via M-TEX-1/panel (READ-ONLY import); per-pair max abs-sample.

Cycle Disposition

Cycle	Researcher Directive	Worker Action	Auditor Decision
1	Ship the milestone under frozen 3-verdict rubric	Full pipeline; MIXED with per-plugin labels; 9 shadow-ledger rows	VALIDATED
2	Verification-only standby	Honest low-output memo; zero writes; recommend loop termination	COMPLETE ([[BRANCH_COMPLETE]])

Test Surface

Suite	Result
tests/test_vst3_nondeterminism_characterization.py	17/17 PASS (exceeds ≥ 14 minimum; includes AST-grep verifying get_state/save_state/save_preset/load_state/set_state(bytes) NOT called; interpreter guard; no PRNG; N=5 runs per plugin; per-run isolated temp dirs; c33 P1 anchor SHAs unchanged; verdict-JSON rubric_hash byte-equal; per-pair metrics finite in expected ranges)
tests/test_integration_cross_branch.py §59 python3 -m long_exposure.tools.promise_check .	11/11 PASS 0 ERRORS (WARNs expected until shadow-ledger merges)

Anchor Preservation

anchor_preservation.json: **153/153 tracked files byte-exact** (c33 dawdreamer_state P1-iterate anchors; c31 palette_probe; c31 palette; c34 palette_v2; c35 palette_v2_render). Zero AST-forbidden state calls; zero PRNG hits; zero forbidden-module imports. c9 chain / c13 pipeline / c15 i4_stratified.py / c22 stability harness / c26-30 utilities NOT imported.

Ledger Events (9 Shadow Rows Under -clone-2 Suffix)

Six named + two housekeeping + one launched (cycle 1); cycle 2 zero:

1. _run/cycle_36_launched-clone-2 (status: validated per c35 Branch C codified convention)
2. _plan/vst3_nondeterminism_characterization_rubric_frozen-clone-2
3. _infra/egress-probe-cycle-36-clone-2
4. M-DAW-SPIKE-1/vst3-render-nondeterminism-characterization (in-progress; M-* unsuffixed per c32)
5. M-DAW-SPIKE-1/vst3-render-nondeterminism-characterization (validated verdict roll-up, MIXED)
6. _run/cycle_36_closed-clone-2
7. _archive/cycle-36-scratch-clone-2
8. _infra/adopt-cycle36-tests-clone-2
9. (per audit tally, 9-row total)

_manager/vst3-render-nondeterminism-anti-pattern-candidate-clone-2 **correctly WITHHELD** under MIXED verdict; c37 owns whether to elevate it if the rubric contradiction resolves toward STRUCTURAL_DRIFT. No M-EAR-1/* or M-GEN-1/* events emitted.

Ledger state: 572 events (563 + 9 emitted this branch).

MODERATE Findings (Handoff to c37, Not Corrections Owed This Branch)

1. **Report §1 wording:** measurement-table row line 34 says “both plugins”; the frozen rubric says “either plugin”. One-line fix owed before c37 reads §1. Not a rework blocker.
2. **Rubric internal contradiction:** doc §98 STRUCTURAL_DRIFT mechanical rule (“ANY of the following ... for EITHER plugin”) vs §118-121 MIXED example (“one plugin passes SMALL while the other passes STRUCTURAL”). Doc line 71 forbids auditor override; preserved per-plugin label in verdict.json means signal is not lost. First rubric-authoring contradiction observed in the campaign.

3. **Hydration symbol mismatch:** brief cited `set_parameters_from_p1_dict` on `scripts.dawdreader_state.probe_p1_iterate_parameters`; symbol does not exist. Worker inlined `c35 render_stem_v2.py:render_dawdreader_vst3_once` hydration loop byte-verbatim; mechanism preserved; `anchor_preservation` confirms `c35` unchanged.

Anti-Patterns Locked (6-Count Stable Including c35 A #6)

`get_state`, `save_state`, `save_preset`, `load_state`, `set_state(bytes)` remain AST-forbidden — enforced by test with 0 hits. `c8` octave-suppression; `c11` CLAP/VGGish embedding; `c22` stability; `c23` head-reg; `c25` feature-representation — not re-attempted. **c35 Branch A VST3-nondeterminism finding survives as fair characterization**, not a re-attempt.

Surge XT drift is inside the VST3 binary, upstream of the P1 API. `c33` P1 hydration surface (100% param coverage) is insufficient to eliminate it; any future Surge XT bisection is narrowed to internal state axes (LFO seeds, voice allocation, envelope phase).

State-Machine Discipline (c29 Lemma Respected)

M-DAW-SPIKE-1/vst3-render-nondeterminism-characterization is a peer sub-milestone under M-DAW-SPIKE-1. NOT a child of terminal-validated M-DAW-SPIKE-1/{palette-schema-v2, palette-schema-v2-hydration-render, palette-instrument-determinism, dawdreader-state-extraction-workaround}.

Merge Disposition

Merge report on disk at `/home/user/music-gen-instance/fork-87da4f517029/clone-2/merge_report.md` for root conductor pickup. Nine shadow-ledger rows queued for `concat_clone_ledgers`; zero cross-clone collisions under `c32 -clone-2` suffixes. Cycle 2 contributes zero additional rows.

Standing Constraints (Unchanged)

- α pinned at 0.7469387071101908.
- SHA-256 tiebreak; no PRNG (AST-verified); no `sidecar_nonfactor` imports.
- Interpreter guard `assert sys.executable == '/usr/bin/python3'` on every new script.
- Read-only anchors preserved (see Anchor Preservation).
- Rated audio egress-blocked at *.googlevideo.com (unchanged 403). M-EAR-1 armed-not-fired posture holds.
- Ledger hygiene: narrative field; `run_id="run-2026-08-28T040704Z"`; nested confidence:{level,rationale,assessment}; `UUID5 content-hash event_id`.

Cycle-37 Handoff (Enumerated in Merge Report + Audit)

Recommended (low risk): adopt per-plugin activation stance from report §9. Any of:

1. **A — Dexed-only strict-SMALL tolerance-gate** (primary; Dexed's byte-adjacent-tolerance is legible under both rubric readings).
2. **B — Surge XT internal-state bisection** (deferred; LFO seed / voice allocation / envelope phase).
3. **C — envelope-only both-plugin tolerance-gate** (higher risk; needs a fresh frozen rubric).

Alternative (higher-effort): formally supersede the rubric doc with a v2 resolving the STRUCTURAL “either” vs MIXED-example contradiction. Requires a fresh pre-registration cycle.

Never re-open c31/c35 A anti-patterns: `get_state`, `save_state`, `save_preset`, `load_state`, `set_state(bytes)` remain AST-forbidden. Surge XT drift is upstream of the P1 API; do not attempt to eliminate it via state extraction.

Report §1 wording fix: one-line “both” → “either” edit before c37 reads §1.

Cycle-Bounding for c37 palette-v3 Decision

The palette-v3 VST3 activation decision does NOT need to answer the Surge XT bisection question first. The c36 Branch C answer (MIXED with per-plugin labels) is sufficient input:

- **Dexed** is a viable candidate for tolerance-gated activation.
- **Surge XT** is not viable for byte-determinism; if used, needs a tolerance-gate rubric.

The c37 fan-out A/B/C choice above operates entirely on this input.

Cumulative Progress

M-DAW-SPIKE-1 line (post-c36 Branch C):

Cycle	Milestone	Verdict
c31	palette-assignment-schema	validated
c31	palette-instrument-determinism	validated (sfizz GREEN; Surge XT + Dexed STILL_GAP)
c33	dawdreamer-state-extraction-workaround	WORKAROUND_FOUND (P1 winning)
c34	palette-schema-v2	SCHEMA_V2_LANDS (four-times-VALIDATED)
c35	palette-schema-v2-hydration-render	RENDER_FAILS (VST3-binary-internal nondeterminism, fair c31 STILL_GAP extension)
c36 (this)	vst3-render-nondeterminism-characterization	MIXED (Surge XT STRUCTURAL / Dexed SMALL, per-plugin labels preserved)

Pattern durability: eight consecutive cycles of rubric-first pre-registration discipline (c26-c30 mechanism probes + c31/c32/c33/c34/c35/c36 palette + infra + fanout). **First rubric-authoring contradiction of the campaign observed this cycle;** soft addition to c37+ discipline: cross-check the rubric’s mechanical rule against its own worked examples before freezing the SHA. Not a hard gate — pre-registration lock already caught it in-audit.

VST3 palette route is now empirically per-plugin heterogeneous: Dexed byte-adjacent-tolerant at float32 epsilon; Surge XT drifts ~10% RMS despite BLAS pins + 100% P1 hydration. c34 palette_v2 schema unaffected.

c29 state-machine lemma respected: peer sub-milestone; ledger topology stays a DAG.

c32 fanout-namespace convention held under c33 harness-clone-namespace-guard: infra families -clone-2-suffixed, substantive M-* unsuffixed.

M-EAR-1 armed-harness Path B: dormant/armed pending audio-egress unblock (still 403; retry per policy is non-blocking). **Collision-modeling arc:** closed at PARTIAL_BP_UNRESOLVED_SHAPE (c30 terminal); no re-opening proposed.

Standby-termination discipline reinforced: worker's own low-output signal fired at cycle 2, aligning with the auditor's no-null-cycle rule. The fanout-harness auto-termination-on-N-consecutive-standby heuristic (carried forward from multiple prior branches) continues to accrue evidence.

[END OUTPUT]